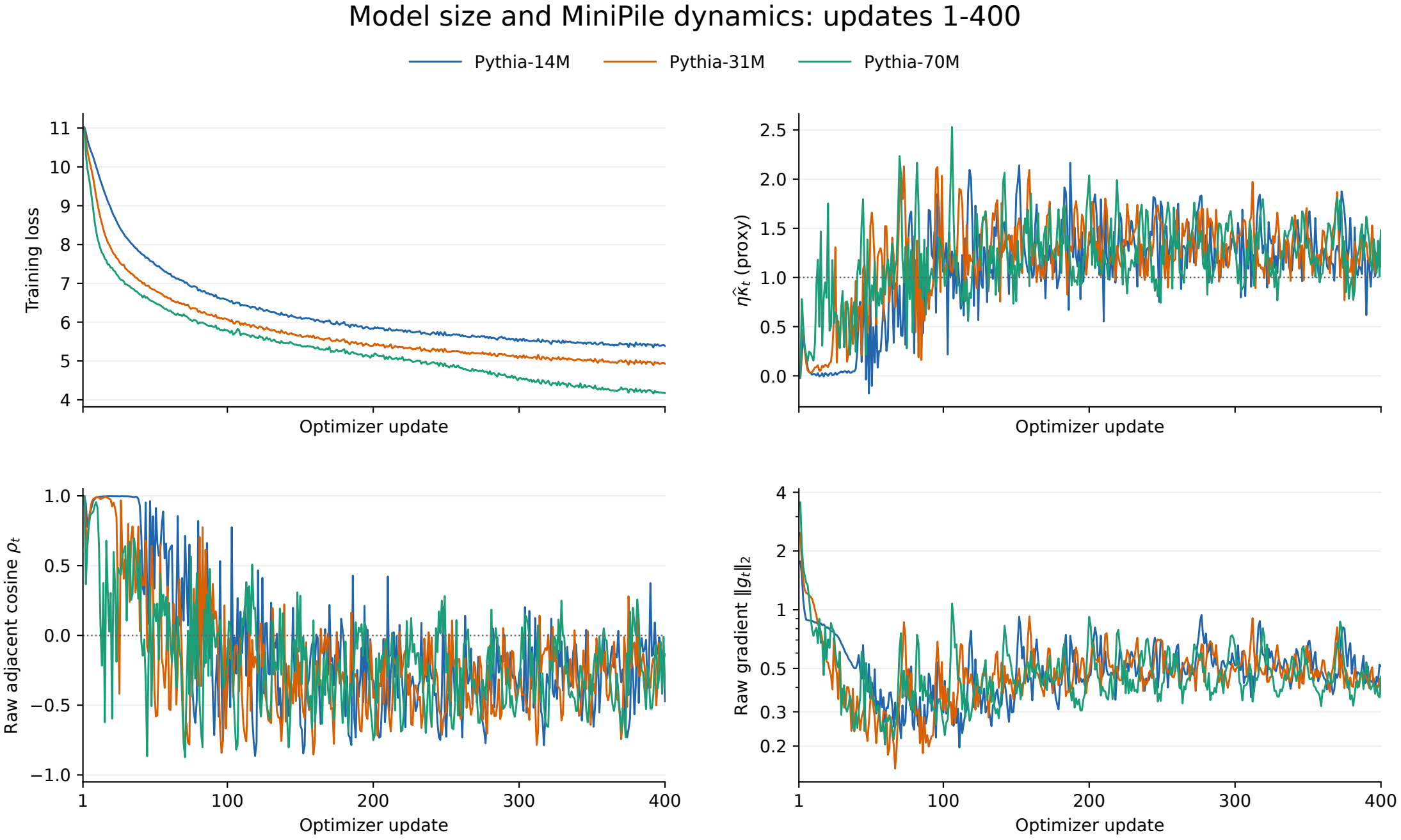

Model size and MiniPile dynamics: updates 1-400
Pythia-14M
Pythia-31M
Pythia-70M
Training loss
Optimizer update
$\hat{\eta}\hat{\kappa}_t$ (proxy)
Optimizer update
Raw adjacent cosine ρ_t
Optimizer update
Raw gradient $\|g_t\|_2$
Optimizer update
All variables unsmoothed; every recorded update 1-400. Seed 1235; solid colours identify model size.
First pair undefined. Norm axis: log Y. AdamW eta*kappa is a gradient-change proxy.